perpendicular distance (d) are constants, the angular momentum of the particle is also constant. Hence, the angular momentum is associated with bodies with linear motion also. If the straight path of the particle passes through the origin, then the angular momentum is zero, which is also a constant.

5.2.5 Angular Momentum and Angular Velocity

Let us consider a rigid body rotating about a fixed axis. A point mass m in the body will execute a circular motion about the fixed axis as shown in Figure 5.12.

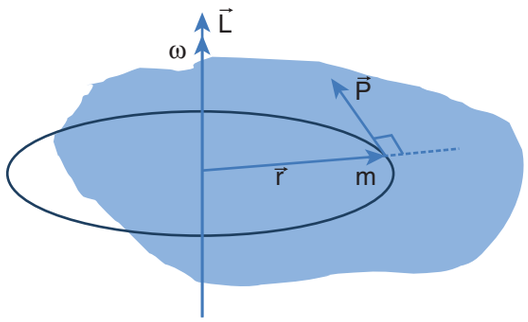


Figure 5.12 Angular momentum and angular velocity

The point mass m is at a distance r from the axis of rotation. Its linear momentum at any instant is tangential to the circular path. Then the angular momentum $\vec{L}$ is perpendicular to $\vec{r}$ and $\vec{p}$. Hence, it is directed along the axis of rotation. The angle θ between $\vec{r}$ and $\vec{p}$ in this case is 90°. The magnitude of the angular momentum L could be written as,

$$L = r\,mv\sin 90^\circ = r\,mv$$

where, v is the linear velocity. The relation between linear velocity v and angular velocity ω in a circular motion is, $v = r\,\omega$. Hence,

$$L = rmr\omega$$

$$L = \left(mr^2\right)\omega \qquad (5.22)$$

The directions of L and ω are along the axis of rotation. The above expression can be written in the vector notation as,

$$\vec{L} = \left(mr^2\right)\vec{\omega} \qquad (5.23)$$

As discussed earlier, the term mr^2 in equations 5.22 and 5.23 is called moment of inertia (I) of the point mass. A rigid body is made up of many such point masses. Hence, the moment of inertia of a rigid body is the sum of moments of inertia of all such individual point masses that constitute the body $\left(I = \sum m_i r_i^2\right)$. Hence, the angular momentum of the rigid body can be written as,

$$\vec{L} = \left(\sum m_i r_i^2\right)\vec{\omega} \qquad (5.24)$$

$$\vec{L} = I\vec{\omega} \qquad (5.25)$$

The study about moment of inertia (I) is reserved for Section 5.4.

5.2.6 Torque and Angular Momentum

We have the expression for magnitude of angular momentum of a rigid body as, $L = I\omega$. The expression for magnitude of torque on a rigid body is, $\tau = I\alpha$

We can further write the expression for torque as,

$$\tau = I\frac{d\omega}{dt} \quad \because \left(\alpha = \frac{d\omega}{dt}\right) \qquad (5.26)$$

UNIT-5(XI-Physics_Vol-1).indd 226 06-01-2022 17:46:51